

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B.Tech (CBCS) Degree Examinations

May 2024

EE2102-1 – ELECTRICAL MACHINES – I
(E&E)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal mark.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- Among the following shapes of the cross-section of the transformer core, the one that leads to minimum leakage flux is
 A) Square
 B) Two-stepped
 C) Rectangular
 D) Three-stepped
- A transformer has negative voltage regulation only when its load power factor is
 A) Zero
 B) Leading
 C) Unity
 D) Lagging
- At light loads, transformer efficiency is low because
 A) Copper loss is high
 B) Transformer heats up more
 C) Input power is high
 D) Fixed loss is high in proportion to the output
- Total power required for Sumpner's test is given by
 A) Double the iron loss of an individual transformer
 B) Iron loss and Copper loss of an individual transformer
 C) Double the Copper loss of an individual transformer
 D) Double the iron loss and double the Copper loss of an individual transformer
- Which of the following three-phase transformer connections has NO path for the flow of third harmonic currents?
 A) Star- star
 B) Star- delta
 C) Delta - delta
 D) Delta - delta
- If a load with unity power factor is supplied from a V-V bank, then out of the two transformers,
 A) One will have a leading power factor while the other lagging power factor
 B) Both will have unity power factor
 C) Both will have lagging power factor
 D) Both will have leading power factor
- The benefit of Copper saving in an autotransformer as compared to an ordinary transformer would be maximum if the turns ratio K is
 A) Far greater than unity
 B) Around unity
 C) Far less than unity
 D) Either far greater than unity or far less than unity
- Among the following three-phase transformer connection configurations, which one requires least insulation for a given pair of primary and secondary line voltages?
 A) Star - delta
 B) Delta - delta
 C) Star- star
 D) Delta - star
- In a three-phase induction motor, the relative speed between stator flux and rotor is:
 A) Synchronous speed
 B) Slip speed
 C) Same as rotor speed
 D) Zero
- In a squirrel-cage induction motor, which of the following rotor quantities does not depend on slip?
 A) Reactance
 B) Resistance
 C) Induced emf
 D) Frequency

11. The slip of an induction motor is the ratio of:
 - A) Rotor Copper loss to rotor input
 - B) Stator Copper loss to stator input
 - C) Rotor Copper loss to stator input
 - D) Rotor Copper loss to stator Copper loss
12. During normal running, the core loss in the rotor of a three-phase induction motor is
 - A) Greater than that in the stator
 - B) Lower than that in the stator
 - C) Equal to that in the stator
 - D) May be lower or greater than that in the stator
13. When supply voltage is reduced by one half, the starting torque reduces to:
 - A) $1/2$
 - B) $1/4$
 - C) $1/8$
 - D) $1/16$
14. When a three-phase induction motor is running under no-load condition, the power factor is:
 - A) Unity
 - B) Zero
 - C) Low
 - D) High
15. An induction machine functions as a generator when it is run
 - A) At synchronous speed
 - B) Above synchronous speed
 - C) Below synchronous speed
 - D) In reverse direction
16. In the stable region of torque-slip characteristics of an induction motor,
 - A) Speed increases as torque increases
 - B) Speed decreases linearly as torque increases
 - C) Speed remains constant as torque increases
 - D) Speed decreases quadratically as torque increases
17. As the rotor resistance of a slip-ring induction motor is increased
 - A) Speed decreases
 - B) Speed increases
 - C) Speed remains constant
 - D) Speed may decrease or increase
18. A star-delta starter is equivalent to an autotransformer starter with a transformation ratio of
 - A) 33.3 percent
 - B) 57.7 percent
 - C) 67.7 percent
 - D) 73.2 percent
19. Crawling of an induction motor is caused by harmonic in stator current.
 - A) 3
 - B) 7
 - C) 9
 - D) 15
20. In a double-cage rotor induction motor,
 - A) The outer cage has high resistance and the inner cage has low resistance
 - B) The outer cage has low resistance and the inner cage has high resistance
 - C) The outer and the inner cage both have high resistance
 - D) The outer and the inner cage both have low resistance

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Derive the emf equation of a single-phase transformer.	4	L*2	1	1
	b) A 200 kVA, 1-phase transformer with a voltage ratio of 6350/660 V has: (i) primary winding resistance of 1.56 ohm (ii) secondary winding resistance of 0.016 ohm (iii) primary leakage reactance of 4.67 ohm (iv) secondary leakage reactance of 0.048 ohm. Calculate the equivalent resistance and equivalent leakage reactance of the transformer referred to high voltage winding.	4	L3	1	1,2
	c) Describe the open-delta operation of a three-phase transformer bank. Why is its VA rating only 57.7% of the closed-delta configuration?	8	L4	2	1,2
2.	a) With neat figures, describe open-circuit and short-circuit tests conducted on a single-phase transformer. Why is the core loss in short-circuit test negligible?	10	L4	1	1
	b) Describe the advantages, disadvantages and applications of auto-transformers.	6	L1	2	1
3.	a) A single-phase transformer has an impedance drop of 10% and resistance drop of 5%. Calculate (i) the regulation at full load 0.8 power factor lagging, (ii) the power factor at which regulation is zero, (iii) the power factor at which regulation is maximum.	6	L3	1	1,2
	b) Which winding polarity is preferred in case of a single-phase transformer? Why?	2	L4	1	1,2

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| c) | Write a short note on tertiary winding. What are its applications? | 8 | L1 | 2 | 1 |
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Unit – II

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| 4. a) | Elaborate the production of the rotating magnetic field in a three-phase induction motor. | 5 | L2 | 3 | 1 |
| b) | A 6-pole, 50 Hz, 3-phase induction motor runs on full-load with a slip of 4%. Given that the rotor standstill impedance per phase is $(0.01 + j0.05)$ ohm/phase, calculate the ratio of maximum torque to full-load torque. Also determine the speed at which the maximum torque occurs. | 7 | L3 | 4 | 1,2 |
| c) | List the advantages and disadvantages of a slip-ring induction motor vis-a-vis a squirrel-cage induction motor. | 4 | L2 | 3 | 1 |
| 5. a) | With relevant circuit diagrams, illustrate no-load and blocked-rotor tests conducted on an induction motor. Why is the no-load current drawn by an induction motor higher as compared to that of a transformer? | 8 | L4 | 3 | 1,2 |
| b) | A 3-phase slip-ring induction motor gives a reading of 60 V across slip-rings when at rest with normal stator voltage applied. The rotor is star connected and has an impedance of $(0.8 + j6)$ ohm per phase. Find the rotor current when the machine is: | | | | |
| | (i) At standstill with the slip-rings joined to a star-connected starter with a phase impedance of $(4+j3)$ ohm | | | | |
| | (ii) Running normally with 5% slip. | | | | |
| | (iii) Find the total Copper loss in the rotor circuit including that in the external resistance for case (i) above. | | | | |
| | (iv) Find the total rotor Copper loss for case (ii) above. | 8 | L3 | 4 | 1,2 |
| 6. a) | Describe the operation of an induction generator. State its limitations. | 8 | L2 | 4 | 1 |
| b) | From the rotor equivalent circuit, derive an expression for mechanical load in terms of rotor resistance and slip. | 5 | L2 | 3 | 1 |
| c) | A 6-pole, 50 Hz, 3-phase induction motor running at full-load develops a useful torque of 150 N-m at a rotor frequency of 1.5 Hz. Calculate the shaft power output. | 3 | L3 | 3 | 1,2 |

Unit – III

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| 7. a) | With a neat circuit diagram, describe star-delta starter. Why is it preferred over DOL starter in case of large induction motors? | 8 | L4 | 5 | 1,2 |
| b) | Why is a single-phase induction motor not self-starting? How is it started in practice? | 5 | L2 | 5 | 1 |
| c) | A 4-pole, 50 Hz single-phase induction motor is running at 1440 rpm. What is its slip with respect to forward-rotating field and backward-rotating field? | 3 | L3 | 5 | 1 |
| 8. a) | With neat circuit diagram and phasor diagram, describe the functioning of a capacitor-start, induction-run single-phase induction motor. | 8 | L2 | 5 | 1 |
| b) | A 3-phase, 6-pole, 50-Hz induction motor takes 60 A at full-load speed of 940 r.p.m. and develops a torque of 150 N-m. The starting current at rated voltage is 300 A. What is the starting torque? If a star/delta starter is used, determine the starting torque and starting current. | 8 | L3 | 5 | 1,2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
